

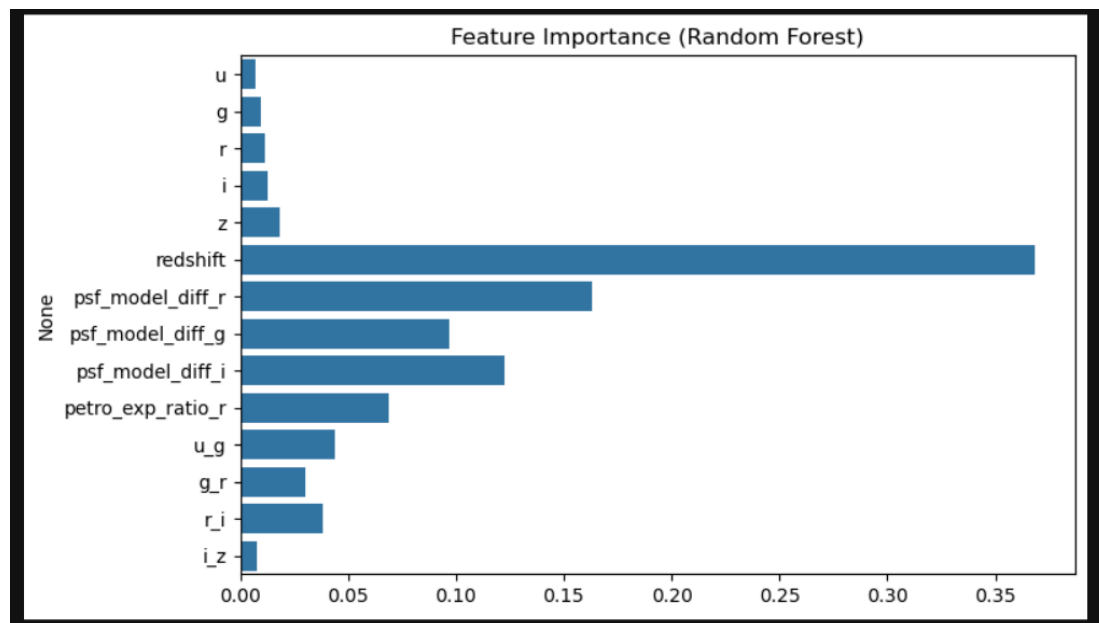
# Results (Color Indices + Morphology)

## Random Forest Classifier (with Redshift)

Classification Report:

| WITH REDSHIFT                |           |        |          |         |  |
|------------------------------|-----------|--------|----------|---------|--|
| Accuracy: 0.9829905726842688 |           |        |          |         |  |
| [[16758 141 98]              |           |        |          |         |  |
| [ 265 3820 0]                |           |        |          |         |  |
| [ 3 0 8722]]                 |           |        |          |         |  |
|                              | precision | recall | f1-score | support |  |
| 0                            | 0.98      | 0.99   | 0.99     | 16997   |  |
| 1                            | 0.96      | 0.94   | 0.95     | 4085    |  |
| 2                            | 0.99      | 1.00   | 0.99     | 8725    |  |
| accuracy                     |           |        | 0.98     | 29807   |  |
| macro avg                    | 0.98      | 0.97   | 0.98     | 29807   |  |
| weighted avg                 | 0.98      | 0.98   | 0.98     | 29807   |  |

Feature Importance:

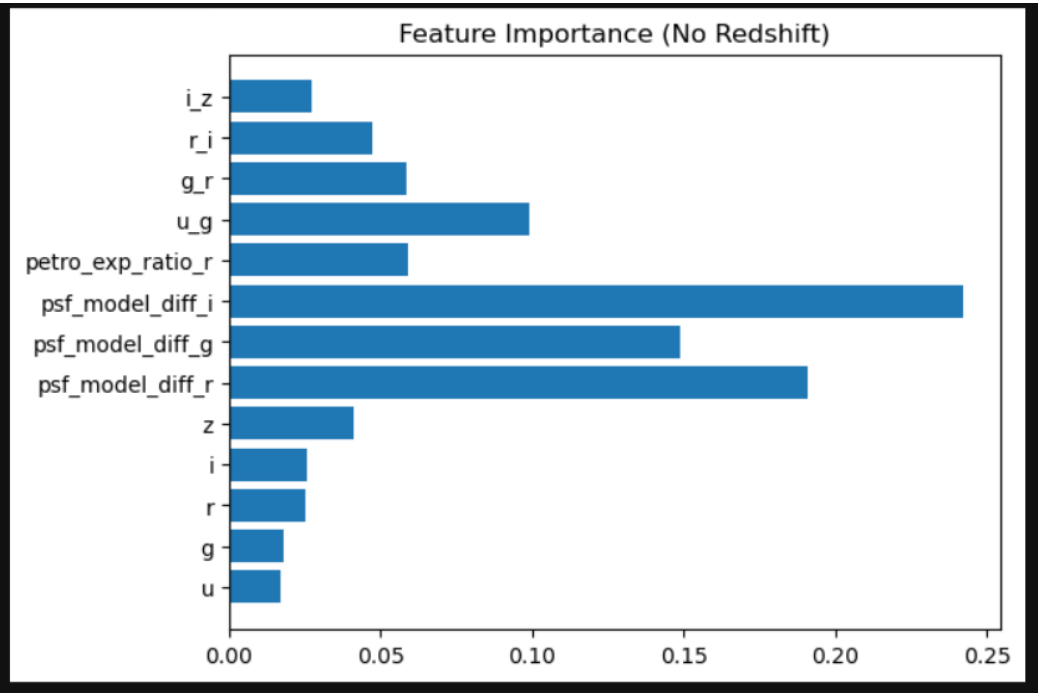


# Random Forest Classifier (Without Redshift)

Classification Report :

|                              |           |        |          |         |  |
|------------------------------|-----------|--------|----------|---------|--|
| WITHOUT REDSHIFT             |           |        |          |         |  |
| Accuracy: 0.9221659341765357 |           |        |          |         |  |
| [[16564 206 227]             |           |        |          |         |  |
| [ 352 2977 756]              |           |        |          |         |  |
| [ 305 474 7946]]             |           |        |          |         |  |
|                              | precision | recall | f1-score | support |  |
| 0                            | 0.96      | 0.97   | 0.97     | 16997   |  |
| 1                            | 0.81      | 0.73   | 0.77     | 4085    |  |
| 2                            | 0.89      | 0.91   | 0.90     | 8725    |  |
| accuracy                     |           |        | 0.92     | 29807   |  |
| macro avg                    | 0.89      | 0.87   | 0.88     | 29807   |  |
| weighted avg                 | 0.92      | 0.92   | 0.92     | 29807   |  |

Feature Importances:



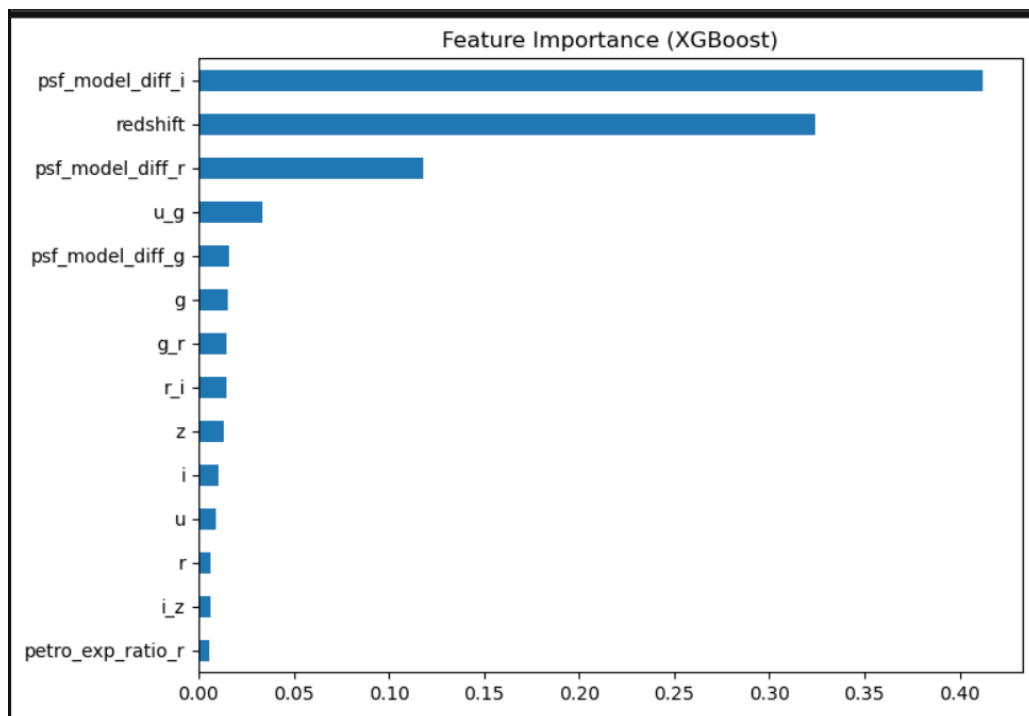
## XGB classifier (with Redshift)

### Classification Report

```
WITH REDSHIFT
Accuracy: 0.9838628510081524
[[16797  124   76]
 [  268 3817   0]
 [   13   0 8712]]
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.98      | 0.99   | 0.99     | 16997   |
| 1            | 0.97      | 0.93   | 0.95     | 4085    |
| 2            | 0.99      | 1.00   | 0.99     | 8725    |
| accuracy     |           |        | 0.98     | 29807   |
| macro avg    | 0.98      | 0.97   | 0.98     | 29807   |
| weighted avg | 0.98      | 0.98   | 0.98     | 29807   |

### Feature Importance:



## XGB classifier (without Redshift)

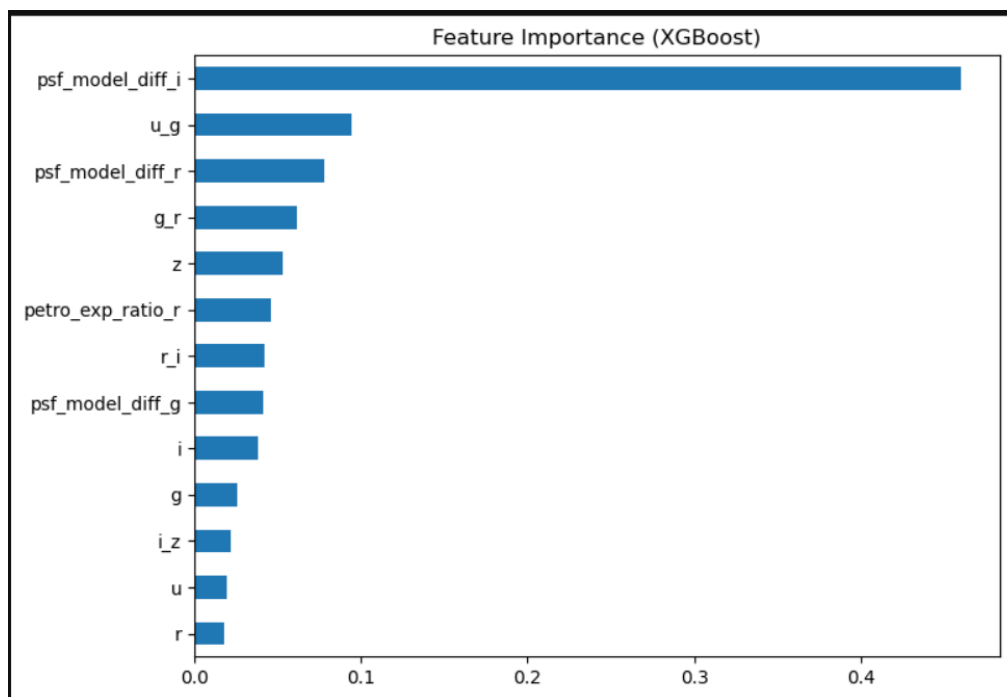
Classification Report:

```
WITHOUT REDSHIFT
Accuracy: 0.9293454557654243
[[16561  201   235]
 [  336 3081   668]
 [   237  429 8059]]

```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.97      | 0.97   | 0.97     | 16997   |
| 1            | 0.83      | 0.75   | 0.79     | 4085    |
| 2            | 0.90      | 0.92   | 0.91     | 8725    |
| accuracy     |           |        | 0.93     | 29807   |
| macro avg    | 0.90      | 0.88   | 0.89     | 29807   |
| weighted avg | 0.93      | 0.93   | 0.93     | 29807   |

Feature Importance :



# FNN Keras (with redshift)

Classification Report:

Test Accuracy: 0.9751065373420715

932/932 1s 1ms/step

Confusion Matrix:

[[16445 409 143]

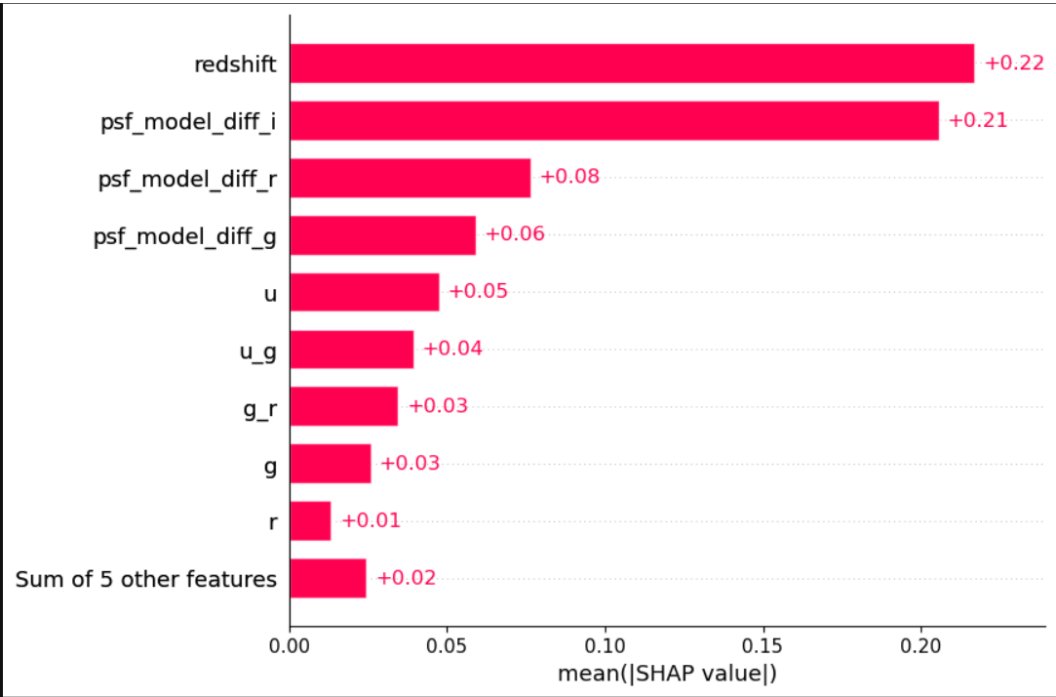
[ 173 3911 1]

[ 15 1 8709]]

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.99      | 0.97   | 0.98     | 16997   |
| 1            | 0.91      | 0.96   | 0.93     | 4085    |
| 2            | 0.98      | 1.00   | 0.99     | 8725    |
| accuracy     |           |        | 0.98     | 29807   |
| macro avg    | 0.96      | 0.97   | 0.97     | 29807   |
| weighted avg | 0.98      | 0.98   | 0.98     | 29807   |

Feature Importance:



## FNN Keras (without Redshift)

Classification Report:

```
Test Accuracy: 0.9141141176223755
932/932 ————— 1s 1ms/step

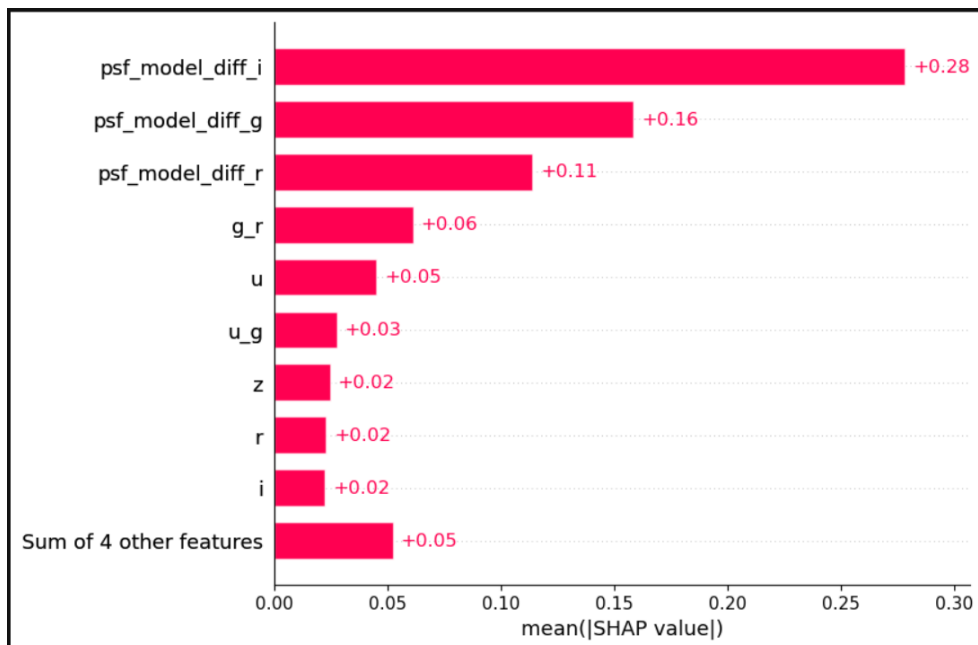
Confusion Matrix:
[[16306  448  243]
 [ 260 3461  364]
 [ 197 1048 7480]]

Classification Report:

```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.97      | 0.96   | 0.97     | 16997   |
| 1            | 0.70      | 0.85   | 0.77     | 4085    |
| 2            | 0.92      | 0.86   | 0.89     | 8725    |
| accuracy     |           |        | 0.91     | 29807   |
| macro avg    | 0.87      | 0.89   | 0.87     | 29807   |
| weighted avg | 0.92      | 0.91   | 0.92     | 29807   |

Feature Importances:



## Stacking (FNN + XGB) (with redshift)

Classification Report:

K-Fold Stacking Accuracy: 0.9848022276646425

Confusion Matrix:

```
[[16792  142   63]
 [  241 3844   0]
 [    7    0 8718]]
```

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.99      | 0.99   | 0.99     | 16997   |
| 1            | 0.96      | 0.94   | 0.95     | 4085    |
| 2            | 0.99      | 1.00   | 1.00     | 8725    |
| accuracy     |           |        | 0.98     | 29807   |
| macro avg    | 0.98      | 0.98   | 0.98     | 29807   |
| weighted avg | 0.98      | 0.98   | 0.98     | 29807   |

Agreement :

XGB Accuracy: 0.9849364243298554  
FNN Accuracy: 0.9783272385681215  
Stacking Accuracy: 0.9848022276646425

Agreement %: 0.9867816284765324  
Accuracy (agree): 0.9881344983510693  
Accuracy (disagree): 0.7360406091370558

## Stacking (FNN + XGB) (without Redshift)

Classification report:

```
K-Fold Stacking Accuracy: 0.9283054316100245
```

```
Confusion Matrix:
```

```
[[16538  225  234]
 [  327 3108  650]
 [  227  474 8024]]
```

```
Classification Report:
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.97      | 0.97   | 0.97     | 16997   |
| 1            | 0.82      | 0.76   | 0.79     | 4085    |
| 2            | 0.90      | 0.92   | 0.91     | 8725    |
| accuracy     |           |        | 0.93     | 29807   |
| macro avg    | 0.89      | 0.88   | 0.89     | 29807   |
| weighted avg | 0.93      | 0.93   | 0.93     | 29807   |

Agreement:

```
XGB Accuracy: 0.9283054316100245
FNN Accuracy: 0.9170329117321434
Stacking Accuracy: 0.9283054316100245
```

```
Agreement %: 0.9474284564028584
Accuracy (agree): 0.9477337110481586
Accuracy (disagree): 0.5781748564135291
```